

- 2 (a) State Newton's law of gravitation.

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 [2]

- (b) One of the basic assumptions of the kinetic theory of gases is that there are no forces exerted between the molecules of the gas except during collisions.

State **two** other basic assumptions of the kinetic theory of gases.

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 [2]

- (c) Hydrogen gas consists of molecules that each have a mass of 3.34×10^{-27} kg. Hydrogen may be considered to be an ideal gas.

A spherical balloon contains 0.0160 mol of hydrogen gas at a temperature of 282 K. At this temperature, the volume of gas in the balloon is $1.87 \times 10^{-4} \text{ m}^3$.

- (i) Determine the pressure of the gas.

pressure = Pa [2]

- (ii) Estimate the average separation of the hydrogen molecules in the gas.

average separation = m [2]

